

Integrating Behavioural and Branding Perspectives to Maximize Green Brand Equity: A Holistic Approach

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ABSTRACT

The aim of this study is to expand the existing understanding of green consumers' behaviour by proposing and testing an integrated conceptual model that explores the influence of consumers' personal concerns for the environment and general attitudes towards green products on brand-related knowledge structures (image and associations) and relationship preferences (trust and brand equity) for green brands. A questionnaire-based survey method was used to collect data using convenience sampling. One hundred and ninety-nine usable responses were obtained. A structural equation modelling procedure was used to test the proposed hypotheses. The results confirmed that a positive relationship exists between consumer concern for environmental values and general attitudes towards green products. Both these constructs influence consumers' knowledge structure of a green brand (image and associations). Furthermore, a strong relationship exists between consumers' knowledge structure (image and associations) and their relational preference (trust and brand equity) with green brands. These findings are important for business strategy formulation by providing empirical support for the idea that a firm should invest its resources not only to project its environmentally friendly brands but also to build consumers' concern for environmental values and their attitude towards green products. Copyright © 2016 John Wiley & Sons, Ltd and ERP Environment

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Introduction

GREEN MARKETING IS DEFINED AS ALL-ENCOMPASSING ORGANIZATIONAL ACTIVITIES, FACILITATING ANY EXCHANGES TO satisfy consumer desires in a way that generates minimal negative impact on the natural environment (Park *et al.*, 2010; Stanton and Futrell, 1987). Thus, green marketing serves as a mean to connect sustainability-driven organizations and their brands with environmentally conscious consumers. An

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organization can green itself by opting for firm or product level strategies (Prakash, 2002). Firm level strategies involve adding green value to existing processes and adopting green management practices. The product level strategy involves developing products with minimal negative environmental impact (Prakash, 2002). In many cases firms incorporate all these strategies in their business plans to maximize the return on their pro-environmental business paradigm.

A recent review of the existing environmental marketing literature by Chamorro *et al.* (2009) indicates the presence of six main research streams. Of the six categories of research topics, two streams, green consumers and green communication, explain 50% of the publications in green marketing. The first stream, which we categorized as the *behavioural perspective* (referred to by Chamorro *et al.*, 2009, as *green consumers*), mainly focuses on investigating the relationship between consumer concern for environmental and other consumption values, general attitudes towards green products and eco-friendly consumption behaviour. This stream of research bases its conceptual models on social adoption theory, value-belief-norm theory, theory of reasoned action and other similar hierarchical models of values, beliefs, attitudes and behaviour (Ajzen and Fishbein, 1980; Biswas and Roy, 2015; Jansson, 2011; Lin and Huang, 2012; Maniatis, 2015; Papista and Krystallis, 2013; Testa *et al.*, 2015; Zhao *et al.*, 2014). Developed in the area of environmental psychology and consumer behaviour, this stream of research only provides a generic model of consumer green purchasing behaviour, generally providing empirical evidence in the context of low involvement post purchase behaviour (Jansson, 2011).

The second stream of research, which we categorized as the *branding perspective* (referred to by Chamorro *et al.*, 2009, as *green communication and its subcategory as anatomy of green ads*) is relatively new in green marketing literature, but is gaining momentum as it tries to explain the impact of green branding initiatives on consumer attitudes and behaviour towards these brands (Bekk *et al.*, 2016; Chen, 2010; Chang and Chen, 2014; Kang and Hur, 2012; Ng *et al.*, 2014; Testa *et al.*, 2015; Tucker *et al.*, 2012). One can argue that this school of research is fundamentally interested in establishing how green brand knowledge structure, which exists in consumers' minds in the form of brand image, awareness, associations and attributes (Esch *et al.*, 2006; Fournier, 1998), influences their responses towards a green brand in shape of their brand trust, satisfaction and long term behavioural commitments (Esch *et al.*, 2006). Associative network memory models, brand signalling theory and social exchange theory form the theoretical underpinnings of this stream of study (Collins and Loftus, 1975; Erdem *et al.*, 2006; Fournier, 1998; Morgan and Hunt, 1994).

The divergence of perspective in these streams of research is evident by the fact that they use completely different theoretical models to investigate green consumption behaviour. The behaviouristic schools explore the issue from a global view of consuming green products, while the branding perspective investigates how specific brands facilitate consumer aspirations towards becoming a green consumer.

Furthermore, despite the significant contribution of both streams of research to green marketing literature, they have their own shortcomings. For example, the green branding stream of research completely ignores the role of personal values and attitudes and other socio-demographic variables in shaping consumer perceptions towards green brands. On the other hand, behaviouristic researchers ignore the role of branding in facilitating consumers' ability to convert their intentions into actions.

It may be that this exclusion of a branding perspective is one of the core reasons that researchers have struggled to establish a strong relationship between environmental values and attitudes and green consumption behaviour. Some studies suggest a very strong relationship while others report weak or moderate relationship between values, attitudes and green consumption (Akehurst *et al.*, 2012; Mostafa, 2007; Tseng and Hung, 2013). This shortcoming might be due to the fact that the 'means to behave' in an eco-friendly manner largely depends on availability of brands that offer green attributes. In our view, the most appropriate approach to address the above shortcomings is by integrating the two schools of thought and moving towards a more holistic model, linking consumer personal values and attitudes with their brand-related cognitive and relationship-oriented constructs. Such a model will provide firms with the much needed empirical evidence that investing in consumers' green education will also pave the way to green brand performance.

This research will strengthen our argument that adopting a holistic green business strategy can help firms to integrate stakeholder's interest, create value on multiple fronts and elevate their role from a political/economic actor to a social actor (Haddock-Fraser and Tourelle, 2010). One way a firm can achieve this objective is by engaging in building environmental collaborations to reduce the cost of educating consumers. For example, recent research

suggests that green claims made through environmental collaborations will not only increase their credibility but will also result in better firm performance (Dangelico and Pontrandolfo, 2015).

Thus the firms which adopt this integrated stakeholder perspective will not only produce and market green brands, but also engage in consumer environmental education. A firm can independently educate consumers or may opt for building environmental collaborations with other stakeholders. Nonetheless, to our knowledge there is little research that has empirically demonstrated potential benefits of investing in consumers' green education to a firm's brand level performance. Nonetheless, some indirect evidence was provided by Tucker *et al.* (2012) in their research on green advertising appeals. They found that consumers' individual personality characteristics influence their involvement with the green advertisement, which subsequently influences their attitude towards the green brand. In simple terms, we do not know if a firm investing in building or reinforcing consumer green values and green attitudes will benefit in the short and/or long term. This research attempts to provide empirical evidence by establishing a positive linkage of consumer general green attitude and environmental values with green perceptions towards a specific brand.

We use the theory of individual self-enhancement (Rogers, 1951), the image congruence hypothesis (Gutman, 1982; Hogg *et al.*, 2000) and the means-end model (Dibley and Baker, 2001) to provide theoretical justifications for the proposed link of consumer concern for environmental values and attitudes with their brand-related cognitive and relationship-oriented constructs. This paper is organized as follows. First it starts with reviewing extant literature, followed by hypothesis development. Sampling, data collection techniques and measurement of variables are discussed in the methodology and measurement section. Results of structural equation modelling (SEM) are given, along with a brief commentary. Finally, conclusions, limitations and implications are discussed.

Theoretical Foundations and Hypotheses

Consumer Values and Attitude Towards Green Products

From a psychological perspective, a value is a belief about one's desired end state (Kim and Chung, 2011; Rokeach, 1973; Schwartz and Blisky, 1987). Values are the most fundamental elements of the human belief system (Kim and Chung, 2011; Vaske and Donnelly, 1999) and play a significant role in moulding individual behaviour and attitudes (Poortinga *et al.*, 2004; Rokeach, 1973). From a social adaptation theory perspective (Bhate 2002; Homer and Kahle, 1988; Kahle, 1983; Leonidou and Leonidou, 2011), values as the most basic abstraction of social cognition serve as a prototype from which specific attitudes and behaviours emerge (Homer and Kahle, 1988; Kahle, 1980). Therefore, in a given context such as green marketing, environmental values will influence attitudes, which will subsequently determine consumer green consumption behaviour. Similarly, value-belief-norm theory also postulates that attitudes and personal values are the most important motivational factors predicting consumers' ecological behaviour (Stern *et al.*, 1999). Other researchers have used the theory of reasoned action, theory of planned behaviour and other hierarchical models of values, beliefs, attitude and behaviour to explain the linkage between consumer values, beliefs, attitudes and intentions towards purchasing green products (Ajzen and Fishbein, 1980; Papista and Krystallis, 2013).

Several past studies reported a positive relationship between consumer values and their behaviour in multiple contexts (Weng and De Run, 2013). In the context of green consumption behaviour, past studies have reported that environmentally conscious consumers are more willing to alter their consumption and purchasing habits in order to contribute towards building a safer environment and a better world (Peattie, 2001; Kim and Chung, 2011). In a recent study, Kim and Chung (2011) reported that personal values such as environment consciousness and appearance consciousness play a more crucial role than other personal values (such as health consciousness) in influencing consumer attitudes towards environmentally friendly products. Therefore, we propose the following hypothesis.

H1: *Consumer environmental values will significantly influence their attitudes towards green products.*

Consumers who are concerned about the environmental consequences of their consumption habits generally tend to hold more positive attitudes towards environmentally friendly brands (Ottman *et al.*, 2006). The

means–end theory postulates that brand attributes lead to benefits, which serve as a means to satisfy personal values (Dibley and Baker, 2001). In the context of the present study, the ‘means–end theory’ provides support for the argument that eco-friendly attributes of a green brand provide environmental benefits to its consumers, which serve as a means to satisfy their concern for environment values. Thus, in accordance with means–end theory, green brands have an environmentally favourable image due to their ability to satisfy consumer concern for environmental values (Gutman, 1982; Dibley and Baker, 2001). Therefore, this study proposes the following hypothesis.

H2: Consumer environmental values are positively related to green brand image.

Attitude is the psychological evaluation of an object, measured by its attributes (Petty *et al.*, 1997). Attitudes are the most consistent explanation of consumers' willingness to actively engage in eco-friendly consumption behaviour (Kanchanapibul *et al.*, 2014). According to value–belief–norm theory, attitudes and personal values are the most important motivational factors predicting consumers' ecological behaviour (Stern *et al.*, 1999). Furthermore, measuring consumer general attitudes towards a product category is fundamentally important because of its tendency to correlate with consumer preferences for a specific brand within that product category (Shamim and Butt, 2013; Kim and Pysarchik, 2000; Walgren *et al.*, 1995). For example, results of recent experimental research suggest that those consumers who scored high on green product attitude also scored high on attitude towards the brand when exposed to an advertisement of green functional appeal (Matthes *et al.*, 2014).

Cretu and Brodie (2007) define brand image as the consumer's visualization of a specific brand, often related to the specific products produced by a company. Consumers are constantly engaged in enhancing their self-concept through symbolic consumption, manifested in acquiring brands that match with a specific dimension of themselves (Grubb and Grathwohl, 1967; Sirgy, 1982). According to the image congruence hypothesis, people's inferences about themselves and others depends on their choice of objects (Belk *et al.*, 1982; Hogg *et al.*, 2000). Furthermore, later expansion in the image congruence hypothesis model proposes that consumption not only serves social symbolism objectives, but also caters to one's private self, where self-worth is achieved by striving to accomplish internal standards (Greenwald, 1989; Hogg *et al.*, 2000).

Thus, in the light of the image congruence hypothesis, one can argue that a consumer who has positive attitudes towards green products will also hold a positive image towards green brands in order to achieve consistency between personal goals and behaviour, either to fulfil his/her own standards or in an effort to generate symbolic recognition.

Therefore, this research proposes the following hypothesis.

H3: Consumer attitude towards green products will positively relate to green brand image.

Based on the proposed relationships in H1, H2 and H3, it is postulated that attitude towards the green products might mediate the relationship between consumer concern for environmental values and green brand image. Therefore we propose the following hypothesis.

H3a: Attitudes towards green products will mediate the relationship between consumer concern for environmental values and green brand image.

The Effects of Green Brand Image on Green Brand Trust

The advocates of green branding argue that a marketer's prime job is to shape positive consumer perceptions towards their green brands (Chen, 2010; Chen and Chang, 2012; First and Khatriwal, 2010; Kang and Hur, 2012; Lee *et al.*, 2010; Mourad and Ahmed, 2012). These researchers advocate that brand image and other brand-related perceptions are primarily created, built and managed by marketers (marginalizing and somewhat ignoring the other sources that might help consumers to build knowledge about a particular brand). They also believe that it is not the green products that consumers purchase, but the brands that develop these products. For example, within a product category such as hybrid cars, some companies were able to enjoy early acceptance of new green products, as in the

case of Toyota Prius, while their competitors (Volkswagen Lupo and Audi A2) struggled (Hartmann and Ibanez, 2006).

In the context of green marketing, a green brand image is defined as 'a whole range of impressions, conceptions and apprehensions towards a brand in the customer's memory, which is correlated to the sustainability and eco-friendly concerns' (Chen, 2010, p. 309; cf. Ng *et al.*, 2014). A successful green brand image only stays in the consumer's mind if the brand is capable of igniting environmentally affirmative consumer emotions, perceptions, attitudes and behaviour (Chen, 2008).

When it comes to building positive attitudes and perceptions towards a green brand, firms face a daunting challenge to build consumer trust in their brands in order to foster long term relationships with their customers. In the context of green marketing, managing and maintaining consumers' green trust is a significant challenge, considering the fact that, in the recent past, claims made by several organizations about the green attributes of their products or services brands have been found to be at best ambiguous, if not overwhelmingly deceptive (Chen and Chang, 2012; Kalafatis *et al.*, 1999; Ng *et al.*, 2014).

Rousseau *et al.* (1998) define trust as 'a psychological state composing the intention to accept vulnerability based on expectations about the intentions or behaviour of another' (p. 395). Chen (2010) defines consumer green trust as a 'willingness to depend on a product, service, or brand based on the belief or expectation resulting from its credibility, benevolence, and ability with regard to its environmental performance' (p. 310).

Nonetheless, those brands with a positive image do influence consumer trust, because a positive brand image is able to reduce consumers' perceived risk of buying a brand (Chang and Chen, 2014). Therefore, the following hypothesis is proposed.

H5: *Green brand image relates positively with green brand trust.*

The Effect of Green Image and Green Trust on Green Brand Equity

Creating an effective green brand image depends on an organization's ability to channel consumers' green or environmentally affirmative emotions, perceptions, attitudes and behaviour towards its brand (Chen, 2008).

Brand image is considered to be one of the most important contributors towards building consumer base brand equity. This has been widely discussed and reported in the past literature (Aaker, 1991; Keller, 1993; Martenson, 2007). Aaker (1991) defined consumer-based brand equity as the 'assets or liabilities which are associated to the name, term, logo or emblem of a brand, whereby it may enhance or depreciate the value that are generated by a good or/and service to the companies' brands or customers' (p. 103). Based on Aaker's (1991) definition of brand equity, Chen (2010) defined green brand equity as 'a set of brand assets and liabilities about green commitment and environmental concerns which are associated with the brand name, symbol and logos that can either elevate or decrease the value given by the eco-friendly goods and services' (p. 313). In this environmentally conscious era, cultivating robust green brand equity can enable companies to achieve competitive advantage by positioning their products differently in the market (Ailawadi and Keller, 2004).

Recent research in the area of green brand equity also reported a positive relationship between green brand image and green brand equity. For example, Chen (2010) finds that green brand image has a positive effect on green brand equity. Similarly, Ng *et al.* (2014) also reported a positive relationship between green brand image and green brand equity.

Therefore, this study also proposes that a green brand image will positively influence consumer-based green brand equity.

H6: *Green brand image relates positively with green brand equity.*

Several past studies such as those by Delgado-Ballester and Munuera-Alemán (2005), Morgan and Hunt (1994) and Jevons and Gabbott (2000) provide empirical evidence suggesting that trust in a brand can exert a positive and direct influence in developing its brand equity. Hence, brand trust acts as one of the most important antecedents of

brand equity (Ambler, 1997). In the context of green marketing, this study also proposes a positive relationship between green brand trust and green brand equity.

H7: *Green brand trust relates positively with green brand equity.*

Based on the proposed relationships in H5, H6, and H7 there is a strong possibility that green brand trust will mediate the relationship between brand image and consumer-based brand equity. Therefore we propose the following hypothesis.

H7a: *Consumer trust in a green brand will mediate the relationship between green brand image and green brand equity.*

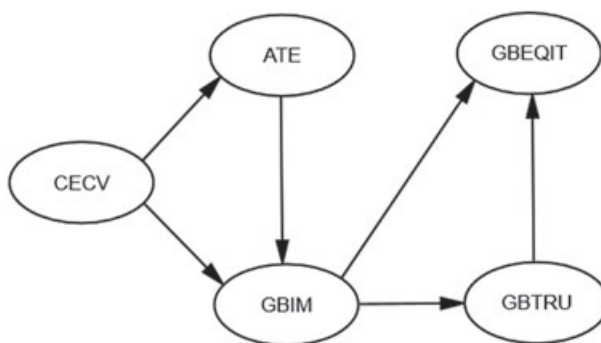
Based on the proposed hypotheses we build a conceptual model (refer to Figure 1). The proposed conceptual model links consumer concern for environmental values with consumer general attitude towards green products and image of recently consumed green brand. The model also proposes that consumers' general attitude towards the green brand will directly influence their specific green-brand-related image. Furthermore, green brand image is a direct predictor of green brand trust and green brand equity.

Research Methodology

Sample and Data Collection

To test the proposed hypotheses, a questionnaire-based survey was conducted among undergraduate and postgraduate students of an international university located in the capital city of Islamabad during the year 2012–13. The potential respondents were required to answer a screening question intended to ensure that they had purchased an environmentally friendly product within the last six months at the time of survey. Only those who answered in the affirmative that they had purchased a green product in last six months were invited to proceed further with the survey questions. The sample comprises undergraduate and postgraduate students, and is based on not only convenience but also the importance of investigating how these young adults perceive environmental issues and its subsequent effect on their consumption patterns.

Young adults form a significant proportion of developing economies, and thus it is not surprising to notice that several researchers have raised the importance of having a sophisticated understanding of this segment of the population (Frank and Chong, 2002; Kanchanapibul *et al.*, 2014). It is also an established fact that the future of environmental protection depends on young consumers taking responsibility for preserving nature by adopting responsible consumption habits (Cini *et al.*, 2012; Maggi *et al.*, 2001). Young consumers are considered more innovative, are socially, environmentally and culturally conscious and demand more rationality compared with other generations



CECV, consumer environmental concern values; ATE, consumer attitude towards green products;
GBIM, green brand image; GBTRU, green brand trust; GBEQIT, green brand equity.

Figure 1. Proposed model

(Hume, 2010; Kanchanapibul *et al.*, 2014; Ottman *et al.*, 2006; Sheahan, 2005; Sullivan and Heitmeyer, 2008). Studies investigating the effect of age on environmentally friendly consumption behaviour have concluded that, as compared with other age groups, younger and well educated consumers are more concerned with issues pertaining to the ecological effects of consumerism (Han *et al.*, 2010; Kanchanapibul *et al.*, 2014; Rowlands *et al.*, 2003; Tseng and Hung, 2013). In addition, by understanding its green consumption behaviour, companies can build more sophisticated marketing strategies to target this important segment of society. Based on this, young adults were selected to test the hypotheses developed in the present study. Three hundred and twenty questionnaires were distributed, out of which 199 usable responses were obtained, representing a 62% response rate.

The data consist of 52% female and 48% male students. Although we did not collect information about respondents' marital status or income, in Pakistan less than 0.5% students get married during their university life (Bagguley and Hussain, 2007). Similarly, like their other South East Asian counterparts Pakistani university students are dependent on their parents' financial support (Bagguley and Hussain, 2007). A majority of students at federal universities in Pakistan represent the lower and upper middle section of society, with family income ranging between one and three thousand dollars per month. Only a small fraction of exceptionally good students from poor backgrounds receive scholarships to study at government higher education institutions.

Research Instrument

This study has adopted previously validated scales to capture consumer concerns for environmental values, and attitude towards green products, green image, green trust and green brand equity. The consumer environmental consciousness value set was measured with three items adopted from the work of Maloney and Ward (1973). Consumer attitudes towards green products were measured with a four-item scale, adopted from Conner *et al.* (1999)

Construct	Measurement items	Standardized Alpha	AVE	CR
Consumer environmental values (CECV)	When I think of the ways industries are polluting the environment, I get frustrated and angry.	0.739	0.748	0.934
	I become irritated when I think about the harm being done to plant and animal life by pollution.	0.769		
	I would be willing to stop buying products from companies guilty of polluting the environment.	0.613		
Attitude towards green products (ATE)	Buying environmental friendly products makes me feel good.	0.836	0.808	0.855
	I love it when the environment friendly brands are available for the product category I purchase.	0.769		0.959
	For most product categories the best buy is usually the environmental friendly brand.	0.673		
	When I buy an environmental friendly brand, I always feel that I am getting a good deal.	0.612		
Green brand image (GBIM)	The brand is professional about environmental reputations	0.771	0.832	0.860
	The brand is well established about environmental concerns.	0.838		0.948
	The brand is trustworthy about environmental promises.	0.771		
Green brand trust (GBTRU)	You feel that this brand environmental commitment is generally reliable.	0.721	0.848	0.877
	You feel that this brand 's environmental argument is generally trustworthy.	0.766		0.966
	The brand environmental concern meets your expectations.	0.767		
	This brand keeps promises and commitments for environmental protection.	0.804		
Green brand equity (GBEQIT)	Even if another brand has the same environmental features as this brand's, you prefer to buy this brand.	0.825	0.821	0.813
	If there is another brand's environmental performance as good as this brand's, you prefer to buy this brand.	0.814		0.928
	If the environmental concern of another brand is not different from that of this brand in any way, it seems smarter to purchase this brand.	0.702		

Table 1. Constructs and their measurement items

and Kim and Chung (2011). Green brand image (five items) and green brand equity (four items) were measured using items from the work of Chen (2010). Green brand trust and green brand equity were measured using scales adopted from Chen (2010).

The data was analysed using Statistical Package for Social Science (SPSS) Version 19.0 and AMOS Version 18. A series of statistical analyses was conducted to test the reliability, validity and proposed hypotheses. Reliability was tested to establish the internal consistency of each construct, and confirmatory factor analysis (CFA) was conducted to confirm the manifestation of variables on their respective constructs. Finally, a full structural equation modelling procedure was used to test the hypothesized relationships between the constructs.

Results

Results of reliability tests showed that the Cronbach alpha score of consumer concern was 0.748 for environmental values (CECV), 0.808 for attitudes towards green products (ATE), 0.832 for green brand image (GBIM), 0.848 for green brand trust (GBTRU) and 0.821 for green brand equity (GBEQIT), indicating acceptable internal consistency for the scales (Nunnally, 1978). To examine the common method bias, Harman's one-factor test for common method variance (CMV) was applied to the data (Podsakoff *et al.*, 2003). Exploratory factor analysis (EFA) on one fixed factor revealed that this factor only explained approximately 38% of variance of the 17 variables. This implies that no CMV exists (Hair *et al.*, 2010). Furthermore, the proposed model is quite complex in its nature and thus inherently reduces potential common method bias.

Confirmatory factor analysis was conducted to establish the convergent and discriminant validity of measurement scales. Measurement items that failed to yield a standard loading of more than 0.50 were deleted from the model (Hair *et al.*, 2010). This resulted in removing two indicators, one item from the consumer green brand image scale and one item from the consumer green brand equity scale. Table 1 reports the AVE (average variance extracted) and CR (composite reliability) for each construct in this research model. All the scales yielded acceptable values for AVE (>0.50) and construct reliability (>0.70), thus establishing convergent reliability (Chen and Chang, 2012; Yap and Khong, 2006). The analysis of measurement model indices indicates that the data have a good fit with the model. Chi square was 127.371 (df = 109, $p = 0.110$), CMIN/df was 1.169, RMSEA was 0.035, IFI was 0.983, TLI was 0.978 and CFI was 0.982.

The test by Fornell and Larcker (1981) was used to establish discriminant validity. Based on this test, the correlation between constructs must not be greater than the square root of the AVE of each construct. Table 2 shows the results of discriminant validity tests on pairs of five constructs. For example, the correlation between GBIM and ATE is 0.430 and the square root of the AVEs of GBIM and ATE were 0.927 and 0.925, respectively. Both AVEs were above the correlation (0.430) between them, implying the presence of discriminant validity. Based on the results presented in Table 2, all constructs have acceptable discriminant validity, as the lower bound square root of AVEs (0.902) was higher than the upper bound values of correlation between green brand trust and green brand

	Square root of AVE	GBTRU	CECV	ATE	GBIM	GBEQIT
GBTRU	0.937	1				
CECV	0.967	0.333**	1			
ATE	0.925	0.493**	0.453**	1		
GBIM	0.927	0.768**	0.550**	0.430**	1	
GBEQIT	0.902	0.711**	0.261**	0.359**	0.508**	1

Table 2. Correlations of the constructs

**Correlation is significant at the 0.05 level (two tailed).

equity (0.768). Thus, one can conclude that the constructs in this proposed research framework (Figure 1) are valid, reliable and distinct from each other (Yap and Khong, 2006).

In accordance with the two-stage modelling, the next step after the confirmation of measurement model is to proceed with the full structural model, using maximum likelihood estimation (MLE), to test the series of hypotheses developed in the proposed framework. The results indicated that the data have an excellent fit with the proposed model. Chi square (χ^2) was 121.619, $p=0.273$ with 113 degrees of freedom. The RMSEA was 0.025, while IFI, TLI and CFI were 0.991, 0.989 and 0.990, respectively. These results implied that the model has absolute and incremental goodness of fit. The χ^2/df (1.076) was between the thresholds of 1 and 2, further indicating model parsimony. A further test was conducted to validate the sample variance–covariance on the structural model. Since there were 199 usable responses, a relatively small sample size, a test for multivariate non-normality was performed. The results show that Mardia's coefficient implied multivariate non-normality on the dataset (Arbuckle, 2009). Subsequently a bootstrap test was performed on the model with 2000 bootstraps. The Bollen–Stine bootstrap p -value was 0.06, indicating that the model has a good fit (Arbuckle, 2009). The overall bootstrapped model fit results were consistently good, with RMSEA reported at 0.027, IFI at 0.990, TLI at 0.987, CFI at 0.989 and χ^2/df at 1.106. The bias-corrected bootstrapped estimates and p -values are presented in Table 3 for comparative purposes. Results show consistency in both p -values and hypotheses, implying that the sample variance covariance on the model had validity (Figure 2).

Furthermore, the residuals of covariance were small, indicating that the model could predict the variance–covariance matrix. All associations were significant ($p < 0.05$) except for hypothesis H6. These results support our initial assumption that there is a possibility of a mediating effect in our model (H3a and H7a). H6 posits a positive relationship between green brand image (GBIM) and green brand equity (GBEQIT), which was not supported in the present study, indicating the possible influence of a mediating variable (green brand trust, GBTRU). Additionally, while consumer environment concern values (CECV) have a positive relationship with green brand equity (GBIM), consumer attitudes towards green products (ATE) can mediate this relationship.

Based on these results, we further tested for the presence of the mediation effect of attitudes towards green products and green brand trust (H3a and H7a) by applying a bootstrapping mediation test procedure in order to establish the possible mediations amongst these associations. A bootstrapping procedure is considered suitable due to its ability to analyse mediation of complex latent constructs (Kenny, 2012). The results of mediation analysis are presented in Table 4, based on the extraction of 2000 bootstrap samples with 95% bias-corrected confidence intervals (Table 5).

Our test for mediation showed that attitudes towards green products (ATE) mediated the association between concern for environmental values (CECV) and green brand image (GBIM). Similarly, green brand trust (GBTRU) mediated the association between green brand image (GBIM) and green brand equity (GBEQIT). Furthermore, mediations between (1) GBIM to GBEQIT (via GBTRU) and (2) CECV to GBIM (via ATE) were significant at $\alpha=0.025\%$. The direct effect of GBIM and GBEQIT was not significant, while the indirect effect was significant, implying that full mediations were present. Meanwhile, the direct and indirect effects of CECV to GBIM were significant, implying that partial mediation was present. As a result, we concluded that GBTRU fully mediated the relationship between GBIM and GBEQIT while ATE partially mediated the relationship and CECV and GBIM.

Parameter	Standardized regression weights	Lower	Upper	p -value
ATE $\leftarrow$ CECV	0.451	0.270	0.617	**
GBIM $\leftarrow$ ATE	0.284	0.095	0.477	**
GBIM $\leftarrow$ CECV	0.401	0.197	0.574	**
GBTRU $\leftarrow$ GBIM	0.782	0.675	0.876	**
GBEQIT $\leftarrow$ GBTRU	0.804	0.526	1.116	**
GBEQIT $\leftarrow$ GBIM	−0.110	−0.446	0.177	0.445

Table 3. Bootstrapped standardized regression weights

**Significant at $\alpha=0.05$.

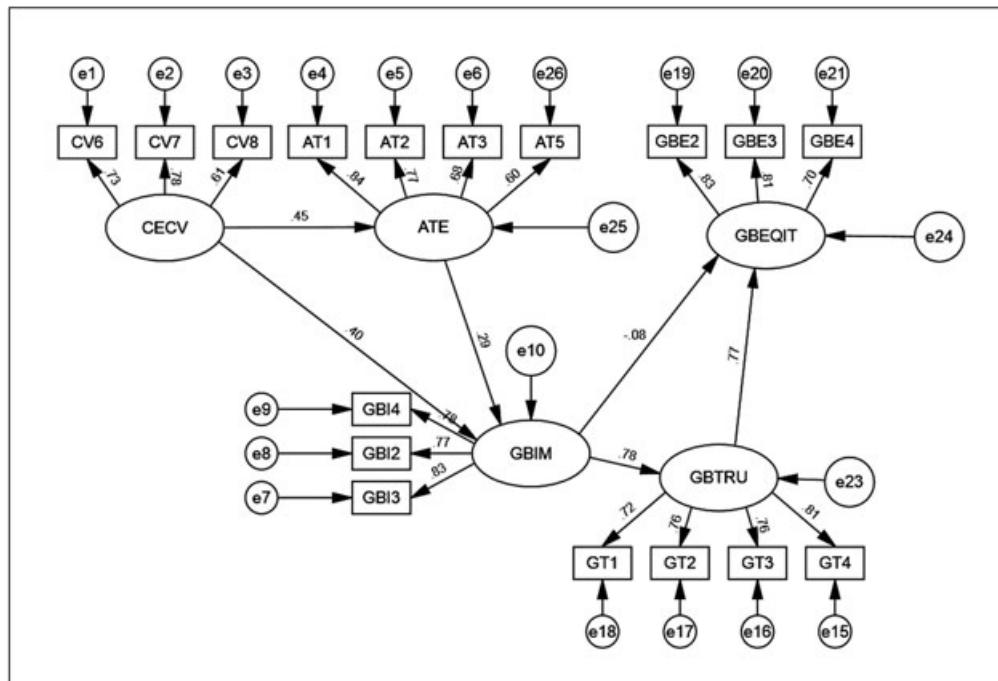


Figure 2. Results of structural model

Discussion and Conclusion

The renewed interest in the area of green marketing has generated new perspectives in the literature from both organizational and consumer perspectives (Chen and Chang, 2012; Lee *et al.*, 2010; Park *et al.*, 2012). In order to expand the existing literature, this study proposed and tested an integrated model examining the effect of consumers' concern for environmental values and their general attitudes towards green products on green brand image and its subsequent influence on green brand trust and green brand equity. In other words, consumer environmental concern values are positively related to brand relationship constructs via a series of mediating variables. This is a significant step forward in the literature of green marketing, as it provides the first empirical evidence that the two schools of research (behaviouristic and branding oriented) are interlinked.

In line with social cognition theory, the results suggest a strong linkage between consumers' concern for environmental values and their general attitude towards green products. This validates several previous studies, which have demonstrated a positive relationship between consumers' concern for environmental values and their buying attitude towards green products (Kim and Chung, 2011; McCarty and Shrum, 1994; Milfont, 2007).

Furthermore, it also provides empirical evidence that these values partially mediate through consumer general attitudes towards green product to influence green brands' image in a positive way, thus providing meaningful support to our argument that there is a positive association between consumers' general attitudes and values and their perception of a specific brand. The positive linkage between consumer general attitude towards green products and brand image suggests that marketers not only need to invest in building positive perceptions of their own green

	Indirect effects <i>p</i> -value	Direct effects <i>p</i> -value	Mediator construct	Mediation
GBIM to GBEQIT	0.001**	0.531	GBTRU	full
CECV to GBIM	0.001**	0.001**	ATE	partial

Table 4. Mediation effects – two tailed significance via bias-corrected (BC) percentile test

**Significant at $\alpha = 0.025$.

Abbreviation	Complete terminology
CECV	consumer environmental concern values
ATE	consumer attitude towards green products
GBIM	green brand image
GBTRU	green brand trust
GBEQIT	green brand equity

Table 5. List of abbreviations

brands, but also need to reinforce consumer concern for environmental values and hence strengthen their global attitudes towards green products. A communication strategy that emphasizes both aspects could be more effective than a unilateral approach. This could be achieved through a two-tiered marketing communication strategy aimed at creating general awareness and building environmental concern values among consumers as well as promoting the greenness of a brand.

In addition, strong and positive relationships were found among brand knowledge structure and consumer relational preferences, such as a positive relationship between green brand image and trust. These results are in support of most of the previous findings that explored relationships between consumers' brand knowledge structure and their relational preferences in a green marketing context (Chen, 2010; Kang and Hur, 2012).

Nonetheless, our results showed no direct relationship between green brand image and green brand equity. This is due to the presence of green brand trust as a fully mediating variable for the relationship between the two constructs. The results of this mediation test showed that trust is an important dimension in building green brand equity, since image alone is insufficient to generate green brand equity unless it is supplemented by consumer trust of that brand. Therefore, marketers interested in pursuing green marketing strategies first need to build into their products green attributes, which are then supported by communication messages, thereby building trust among consumers.

We draw several implications from the results of this study. This study provides empirical evidence that consumer concern for environmental values and general attitudes towards green products are important antecedents of green brand image. Thus, organizations that are involved in the development and promotion of green products can pool their resources together through environmental collaborations to build a national or international promotional campaign that helps strengthen consumer concern for environmental values and general attitudes towards green products. For example, in a recent study, researchers empirically validated the positive impact of a firm's ability to develop and implement environmental collaborations on the firm's market and image performance (Dangelico and Pontrandolfo, 2015). The advertising campaign 'Got Milk' is one example from the past that shows how a particular industry can pool together to build positive public opinion towards a product category (Bower and Mateer, 2008; Hsu and McDonald, 2002). Therefore, we also suggest that, apart from individual investment in promoting their green brands, firms can pool resources to launch a systematic campaign to create awareness of green products in general and to reinforce consumers' environmental values and attitudes towards green products.

Furthermore, when a firm integrates green marketing approaches in its overall business strategy, it stimulates a culture of innovation to incorporate new attributes into products, which when experienced by consumers will encourage trust in them, which in turn can have a positive effect on brand equity. Thus only those organizations that incorporated green marketing initiatives as an integral part of their overall business strategy were successful in attracting environmentally concerned customers (Chamorro and Bañegil, 2006). This is further supported by recent research that provides empirical evidence suggesting a positive relationship between consumer innovativeness and early adoption of products with green attributes (Jansson, 2011). A firm may by positioning itself as a frontrunner for green innovation find it easier to generate a feeling of self-congruence between its brand and early adopters, which are the most important target segment for markets of new products.

These results should be interpreted while considering the limitations of this study. Future studies could explore the implicit motivations for preference for green products using qualitative approaches such as projective techniques or laddering. Based on these methods, new dimensions of why and how consumers favour green brands

can be better understood. A longitudinal study that tracks changes in consumer attitudes and behavior towards green products could also be insightful for marketers and researchers in understanding the impact of environmental concerns and values over the longer term.

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